



Inspiring Excellence

**Project Report: 30-Day Repeat Purchase Prediction from Online  
Retail II**

Summer 2026

CSE 437

Section 5

Group 17

Sahir Jawad Chowdhury

23301212

Github Repository:

<https://github.com/Sahirsmf/cse437-30-Day-Repeat-Purchase-Prediction-from-Online-Retail-II-17.git>

03.09.2026

# Summary

This project uses the UCI Online Retail II transaction dataset to predict whether an identified customer will make another purchase within 30 days of a monthly prediction date. The raw data contain 1,067,371 transaction records from December 2009 to December 2011. After duplicate removal and purchase-specific filtering, 793,609 customer-identifiable purchase transactions were used to construct 61,852 customer-period observations. The binary target has 27.44% positive cases overall. Leakage-safe behavioral features were built from the preceding 180 days, while the next 30 days were used only to define the target. Logistic Regression and Random Forest were compared with time-based train/validation/test splits and F1 as the primary metric. On the held-out test period, Random Forest achieved the highest F1-score, 0.5455, narrowly ahead of Logistic Regression at 0.5439; its PR-AUC was also slightly higher, 0.6047 versus 0.6000. Behavioral variables such as total spend, recency, quantity and purchase frequency were much more important than country. The evidence supports moderate repeat-purchase prediction, but does not establish that feature selection improves performance.

## 1. Problem and Dataset

### 1.1 Problem statement

The task is binary classification at the customer × prediction-date level. Each monthly example uses the preceding 180 days of purchase history to predict whether the customer makes at least one purchase in the following 30 days. This can support customer-retention prioritization while keeping predictors strictly pre-outcome.

### 1.2 Dataset

UCI Online Retail II; 1,067,371 raw rows, 9 columns, 5,942 identified customers, 43 countries, and dates from 2009-12-01 to 2011-12-09. The notebook combines the two Excel sheets and adds SourceSheet. UCI lists the dataset under CC BY 4.0 (Chen, 2012). Source: <https://archive.ics.uci.edu/dataset/502/online%2Bretail>

### 1.3 Target variable

return\_30d is binary: 0 = no purchase in the next 30 days; 1 = at least one purchase. The 61,852 customer-period observations contain 72.56% negatives and 27.44% positives. The held-out test set has 11,662 observations and 32.19% positives.

### 1.4 Three questions

- Q1. Can we predict whether a customer will purchase again within 30 days?
- Q2. Which customer behaviors are most important for predicting repeat purchases?
- Q3. How does feature selection affect model performance?

## 2. Data Handling and Preprocessing

### 2.1 Data quality audit

| Issue                 | Count   | % raw  |
|-----------------------|---------|--------|
| Exact duplicates      | 12,133  | 1.14%  |
| Negative quantities   | 22,950  | 2.15%  |
| Zero prices           | 6,202   | 0.58%  |
| Negative prices       | 5       | ~0.00% |
| Missing Customer ID   | 243,007 | 22.77% |
| Missing Description   | 4,382   | 0.41%  |
| Cancellation invoices | 19,494  | 1.83%  |

### 2.2–2.4 Missing values, outliers, transformation and scaling

Customer ID was not imputed because the target is customer-level; unidentified rows were retained for audit continuity but

excluded from customer modeling. Description was not needed by the final feature set. Outliers were flagged using  $1.5 \times \text{IQR}$ : 51,910 Quantity, 66,736 Price and 64,720 TotalPrice values. They were retained because large legitimate orders are plausible. Numeric modeling features use median imputation and StandardScaler; Country uses most-frequent imputation and one-hot encoding. These transformations are fitted inside the modeling pipeline after the temporal split.

## 2.5 Before and after

| Stage                  | Rows      | Columns |
|------------------------|-----------|---------|
| Raw combined           | 1,067,371 | 9       |
| After duplicates       | 1,055,238 | 9       |
| After date handling    | 1,055,238 | 10      |
| Purchase-only modeling | 793,609   | 11      |

## 3. Statistical Analysis

### 3.1 Descriptive statistics

Raw Quantity has mean 9.94, median 3 and skewness 3.996. Price has mean 4.65, median 2.10 and skewness -69.165. TotalPrice has median 9.90, mean 18.07 and maximum 168,469.60, showing heavy tails.

### 3.2 Relationships

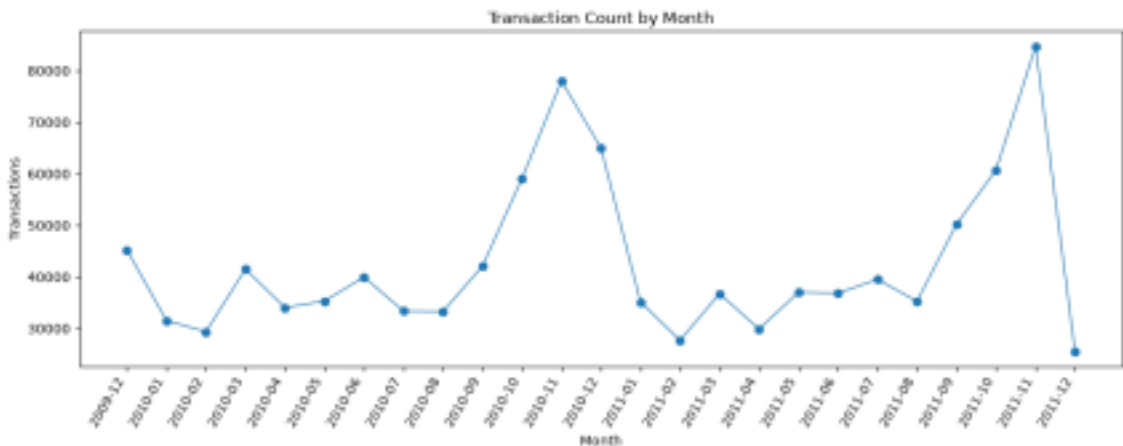


Figure 1. Monthly transaction count.

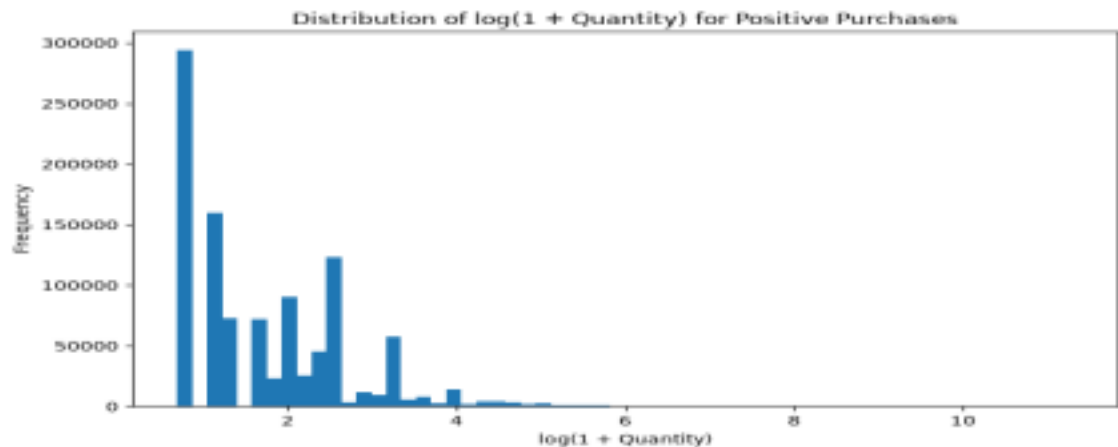


Figure 2. Log(1 + Quantity) distribution for positive purchases.

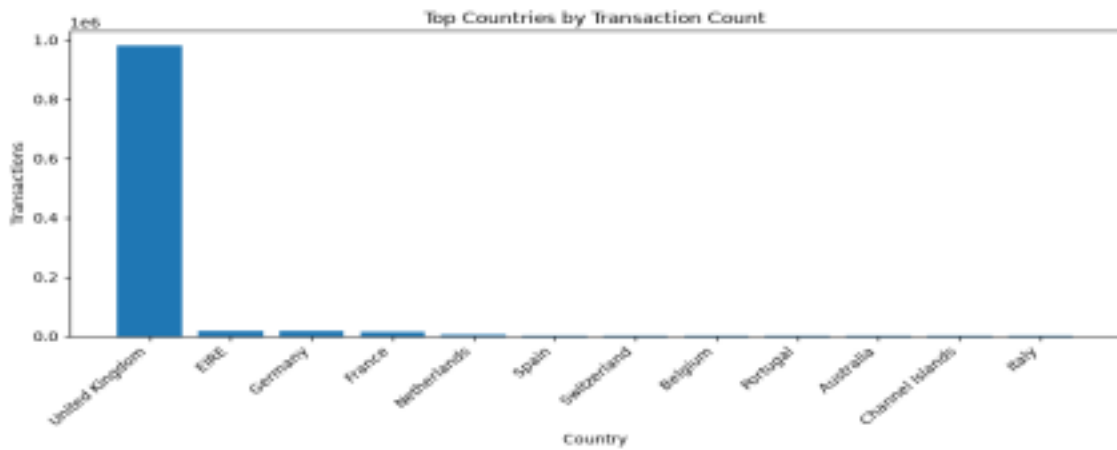


Figure 3. Top countries by transaction count.

### 3.3 What the data says so far

Repeat purchase is a minority outcome; frequency and customer longevity show strong univariate signal; spending and quantity also matter; AverageOrderValue and AvgItemsPerTransaction have weak F-test evidence; country is much less important than behavioral variables in the Random Forest.

## 4. Feature Engineering

### 4.1 Derived features

Twenty-two monthly prediction dates were used. Ten numeric features summarize 180-day recency, frequency, spending, quantity, product diversity and activity, plus the customer's most recently observed Country. The target is defined only from the following 30 days, preventing future leakage.

### 4.2–4.4 Dimensionality reduction, selection and final set

PCA on standardized numeric features needs 6 components to exceed 90% variance (92.92%). PCA was not retained. Training-only  $f_{\text{classif}}$  ranking places ActiveMonths, PurchaseDays and TransactionCount highest. However, the notebooks do not specify a final K/threshold or evaluate a selected-feature model. The final models therefore use all ten numeric features plus Country.

## 5. Modeling and Validation

### 5.1–5.4 Validation, baseline, model families and metrics

The split is 70% train (41,758), 15% validation (8,432), 15% test (11,662), ordered by PredictionDate. Tuning uses 4-fold TimeSeriesSplit on training data only. The majority baseline has accuracy 0.6781 but precision, recall and F1 of 0.0. Logistic Regression provides an interpretable linear model; Random Forest captures nonlinearities and interactions. F1 is the primary metric because the positive class is not the majority; ROC-AUC and PR-AUC are secondary measures.

## 6. Hyperparameter Tuning

### 6.1–6.3 Search, method and results

| Model               | Search space                                                                | Best configuration | Best CV F1 |
|---------------------|-----------------------------------------------------------------------------|--------------------|------------|
| Logistic Regression | C: .01/.1/1/10; solver: liblinear/lbfgs                                     | C=10, liblinear    | 0.5095     |
| Random Forest       | n_estimators: 200/400; depth: None/8/16; leaf: 1/3/8; max_features: 0.5/1.0 |                    | 0.5068     |

There were 8 LR and 36 RF grid candidates, scored by 4-fold temporal CV. The executed notebook reports the best settings and scores but does not display the full candidate-by-candidate trend, so that trend is not claimed here.

# 7. Results, Visualization and Error Analysis

## 7.1 Test set performance

| Model    | Accuracy | Precision | Recall | F1     | ROC-AUC | PR-AUC |
|----------|----------|-----------|--------|--------|---------|--------|
| Baseline | 0.6781   | 0.0000    | 0.0000 | 0.0000 | —       | —      |

| Model               | Accuracy | Precision | Recall | F1     | ROC-AUC | PR-AUC |
|---------------------|----------|-----------|--------|--------|---------|--------|
| Logistic Regression | 0.6983   | 0.5298    | 0.5589 | 0.5439 | 0.7193  | 0.6000 |
| Random Forest       | 0.7059   | 0.5427    | 0.5482 | 0.5455 | 0.7206  | 0.6047 |

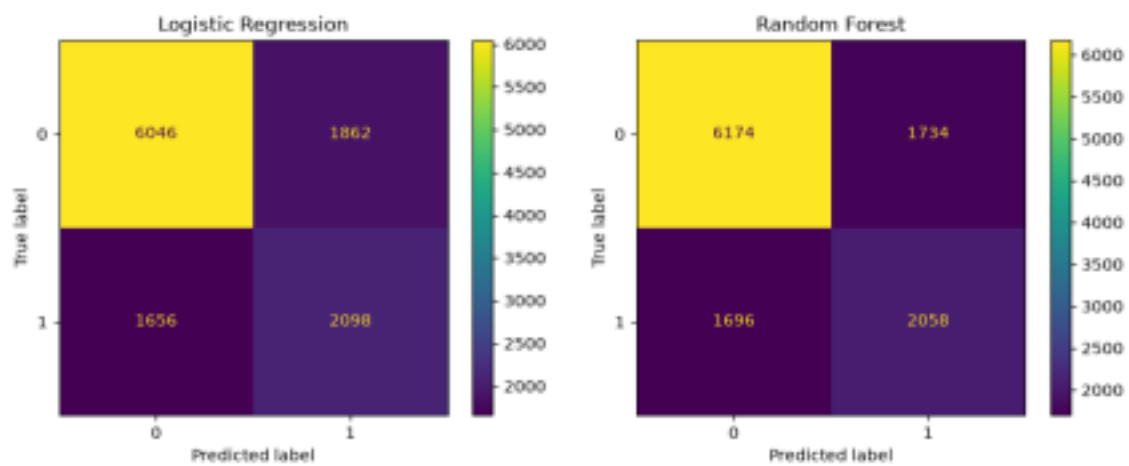


Figure 4. Confusion matrices; RF: TN 6,174, FP 1,734, FN 1,696, TP 2,058.

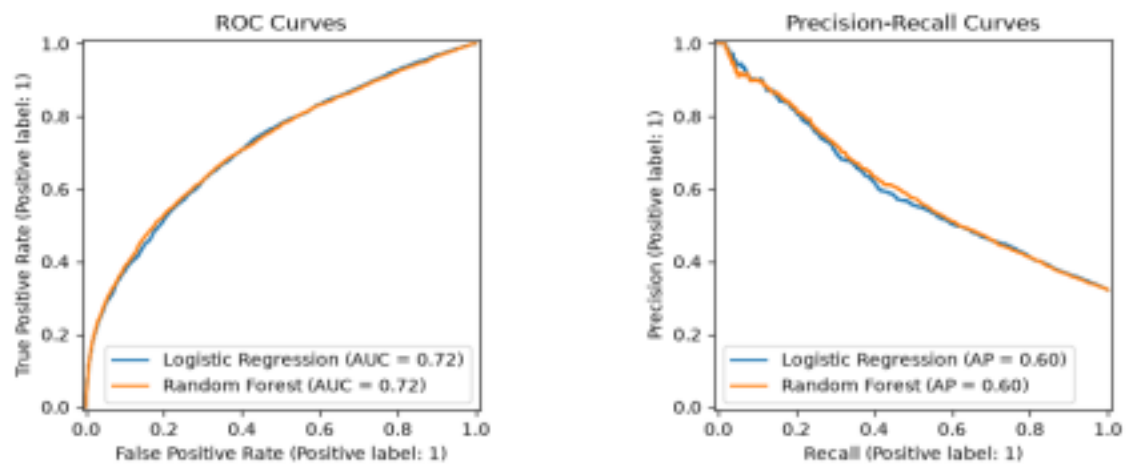


Figure 5. ROC and precision-recall curves.

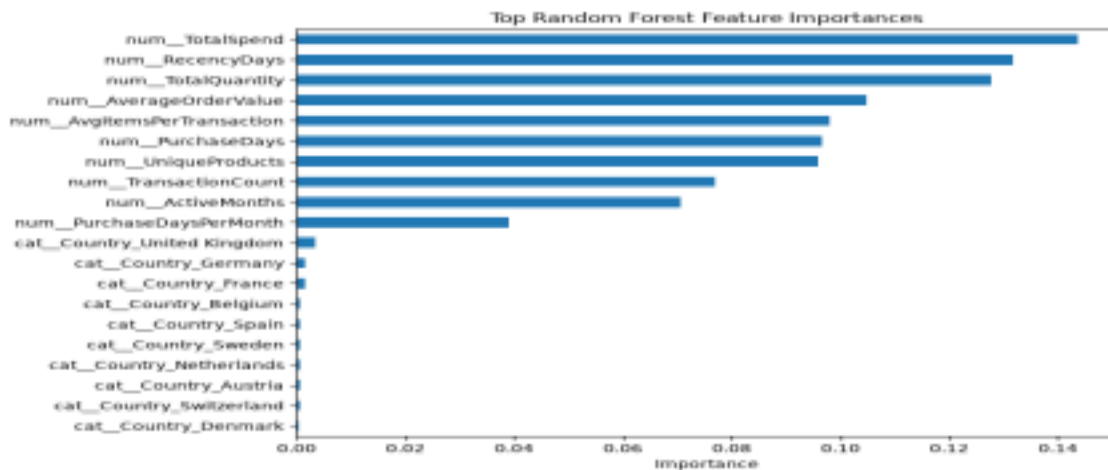


Figure 6. Random Forest feature importance.

### 7.3 Error analysis

Random Forest has 1,734 false positives and 1,696 false negatives. Example FP: customer 17961 on 2011-10-01, true 0, RF probability 0.9609. Example FN: customer 15299 on 2011-10-01, true 1, RF probability 0.0613. The supplied notebook does not print their historical feature profiles, so a feature-based explanation of these failures cannot be made without inventing evidence.

### 7.4 Answers to the three questions

Q1: Yes, moderately; RF F1 = 0.5455 versus baseline F1 = 0.0.

Q2: Behavioral activity dominates, especially spend, recency, quantity and frequency-related measures.

Q3: Feature selection highlighted key behavioral variables such as purchase frequency, engagement, recency, and product breadth. However, no direct comparison with full-feature models was available, so its impact on F1-score cannot be confirmed. PCA retained six components explaining 92.9% of variance, but was excluded from the final pipeline to preserve feature interpretability.

## 8. Limitations and Next Steps

The main limitations are moderate predictive performance, temporal generalization, substantial missing Customer IDs, retained extreme values, incomplete feature-selection comparison, incomplete error-profile inspection, and missing full grid-search trend output. Next steps are to evaluate explicit selected-feature models, tune the decision threshold using validation data, assess probability calibration, and perform systematic subgroup and feature-profile analysis of errors.

## 9. Contributions

| Member                | Student ID | Contribution    |
|-----------------------|------------|-----------------|
| Sahir Jawad Chowdhury | 23301212   | Did everything. |

## References

Chen, D. (2012). Online Retail II [Dataset]. UCI Machine Learning Repository. DOI: 10.24432/C5CG6D.  
<https://archive.ics.uci.edu/dataset/502/online%2Bretail>

Scikit-learn components used in the notebooks: Pipeline, ColumnTransformer, StandardScaler, OneHotEncoder, LogisticRegression, RandomForestClassifier, TimeSeriesSplit, GridSearchCV, SelectKBest, PCA and classification metrics.

Five supplied project notebooks: 01\_data\_audit\_and\_edu.ipynb; 02\_preprocessing.ipynb; 03\_feature\_engineering.ipynb; 04\_modeling\_and\_tuning.ipynb; 05\_evaluation\_and\_error\_analysis.ipynb.

AI assistance declaration: ChatGPT was used to organize notebook evidence into the required report structure, draft prose, and format the report. Numerical results and methodological claims were taken from the supplied notebooks; unsupported gaps were explicitly identified.